

Program : Diploma in Cyber Forensics and Information Security	
Course Code : 5289C	Course Title: Internet and Web Application Security Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- Presenting internet security systems and covering most important parts of cryptography through introducing many cryptography techniques and algorithms.
- Explaining the hash function as an application of cryptography aligning with the concept of message integrity and digital signature authentication.
- Understand the tools for web application vulnerability assessment and defending malwares.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic knowledge in Cryptography		Fundamentals of Cryptography and Information Security	III

Course Outcomes:

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Presenting the most important substitution and transposition techniques in cyber security.	9	Applying
CO2	Utilizing and code developing for encryption algorithms that required to achieve confidentiality key security.	10	Applying
CO3	Building an appropriate encrypting system that designed for specific key size and message length.	10	Applying
CO4	Presenting Tools for Intrusion Detection System, Vulnerability Assessment and defeating malwares	13	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3				
CO3	3		3				
CO4	3		3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Presenting the most important substitution and transposition techniques in cyber security.		
M1.01	Perform encryption, decryption using the following substitution techniques i. Ceaser cipher ii. Playfair cipher iii. Hill Cipher iv. Vigenere cipher	5	Applying
M1.02	Perform encryption and decryption using following transposition techniques i. Rail fence ii. Row & Column Transformation	4	Applying
CO2	Utilizing and code developing for encryption algorithms that required to achieve confidentiality key security.		
M2.01	Apply DES algorithm for practical applications.	3	Applying
M2.02	Apply AES algorithm for practical applications.	3	Applying
M2.03	Implement RSA Algorithm using HTML and JavaScript	4	Applying
	Lab Test – I	1.5	
CO3	Building an appropriate encrypting system that designed for specific key size and message length		
M3.01	Implement the Diffie-Hellman Key Exchange algorithm for a given problem	3	Applying

M3.02	Calculate the message digest of a text using the SHA-1 algorithm.	4	Applying
M3.03	Implement the SIGNATURE SCHEME - Digital Signature Standard..	3	Applying
C04	Presenting Tools for Intrusion Detection System, Vulnerability Assessment, defeating malwares.		
M4.01	Demonstrate intrusion detection system (ids) using any tool eg. Snort or any other s/w	2	Applying
M4.02	Automated Attack and Penetration Tools Exploring N-Stalker, a Vulnerability Assessment Tool	3	Applying
M4.03	Defeating Malware i. Building Trojans ii. Rootkit Hunter	3	Applying
	Open-ended projects	5	
	Lab Test – II	1.5	

Sample Open Ended Projects

(Students can do open ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

- Test website for vulnerabilities and threats. (can use Acunetix web vulnerability scanner (WVS) <http://www.acunetix.com/vulnerability-scanner>)
- Parameter tampering attacks and Cross-site scripting (XSS or CSS)

Text / Reference

T/R	Book Title/Author
T1	Cryptography and Network Security: Principles and Practice, Global Edition, 7/E, William Stallings, Pearson
T2	Padmanabhan T R, Shyamala C and Harini N, "Cryptography and Security", Wiley Publications 2011.

Online Resources

Sl.No	Website Link
1	https://www.tutorialspoint.com